



Module 1 challenge

1st Attempt:

1. A data professional investigates which programming language they can use to perform their coding work most effectively. They want to use a versatile language based on simple syntax. Ultimately, they select Python. What are some other benefits of using Python for coding? Select all that apply.

1 / 1 point

☒ Python is easy to learn.

✔ Correct

☐ Python is a low-level language.

☒ Python has a large and supportive community of users.

✔ Correct

☒ Python has thousands of open source libraries available for use.

✔ Correct

2. Fill in the blank: The open-source web application _____ enables users to share documents containing live code, mathematical formulas, visualizations, and text. It also puts all output into a single document, which is very useful when first learning about programming.

1 / 1 point

☒ Jupyter Notebook

☐ Dot notation

☐ Implicit analysis

☐ Dynamic coding

✔ Correct

3. Fill in the blank: In Python, an _____ is a value associated with an object or class which is referenced by name using dot notation.

1 / 1 point

- ☐ expression
- ☐ integer
- ☐ assignment
- ☒ attribute

✓ Correct

4. In Python, what describes the process of storing a value in a variable?

1 / 1 point

- ☒ Assignment
- ☐ Syntax
- ☐ Expression
- ☐ Conversion

✓ Correct

5. A data professional names a variable in Python, then receives an error message. What might they have included in the name incorrectly? Select all that apply.

1 / 1 point

- ☐ More than one number
- ☒ Reserved functions

✓ Correct

- ☒ Reserved keywords

✓ Correct

- ☐ More than one underscore

6. Which of the following is an effective variable name?

1 / 1 point

- ☒ `able_variable`
- ☐ `able(variable)`
- ☐ `5_able_variables`
- ☐ `_able_int_variable`

✓ Correct

7. In Python, which data type represents numbers that contain decimals?

1 / 1 point

- ☐ Integer
- ☐ String
- ☐ Boolean
- ☒ Float

✔ Correct

8. A data professional performs a calculation involving integers and floats. Python automatically converts the integers to floats. This is an example of what type of conversion?

1 / 1 point

- ☐ Mutable conversion
- ☒ Implicit conversion
- ☐ Explicit conversion
- ☐ Immutable conversion

✔ Correct

9. You're working in a Jupyter notebook. Consider the following execution of cells:

1 point

Cell 1:

```
x = 5
```

Cell 2:

```
x = 10
```

Cell 3:

```
print(x)
```

```
x = 20
```

If you run each cell in order and only one time, what will the output of cell 3 be?

- ☒ 20
- ☐ 5
- ☐ x
- ☐ 10

✘ Incorrect

The output would be **10** because code executes line-by-line and cell-by-cell. In cell 2, **x** is reassigned a value of **10**. The first line in cell 3 is a **print** statement, and at the time of its execution, the value of **x** is still **10**. The value of **x** is reassigned to **20** *after* the **print** statement executes. Refer to the [Variables and datatypes](#) [video](#) to review how variable assignment works in Python.

10. Which of the following are valid statements (i.e., will NOT result in a code error) in Python? Select all that apply.

1 / 1 point

☒ `6 > 3`

✓ Correct

☒ `14 % 2`

✓ Correct

☒ `x == 2`

✓ Correct

☐ `15.56 \ x`

2nd Attempt:

1. A manager explores which programming language their team can use to perform coding work most effectively. They want a language with thousands of open-source libraries for reference. Ultimately, they select Python. What are some other benefits of using Python for coding? Select all that apply.

1 / 1 point

☒ Python is based on simple syntax.

✓ Correct

☒ Python is versatile.

✓ Correct

☒ Python has a large and supportive user community.

✓ Correct

☐ Python is a low-level language.

2. Fill in the blank: The open-source web application _____ enables data professionals to modularize code into cells in order to organize them in sections, as well as collaborate on data projects and integrate code.

1 / 1 point

☐ Implicit analysis

☐ Dot notation

☒ Jupyter Notebook

☐ Dynamic coding

✓ Correct

3. Fill in the blank: In Python, an _____ is a value associated with an object or class which is referenced by name using dot notation.

1 / 1 point

- ☒ attribute
- ☐ integer
- ☐ assignment
- ☐ expression

✓ Correct

4. In Python, what describes the process of storing a value in a variable?

1 / 1 point

- ☐ Syntax
- ☒ Assignment
- ☐ Conversion
- ☐ Expression

✓ Correct

5. A data professional names a variable in Python, then receives an error message. What might they have included in the name incorrectly? Select all that apply.

1 / 1 point

- ☐ More than one number
- ☒ Reserved functions

✓ Correct

- ☐ More than one underscore
- ☒ Reserved keywords

✓ Correct

6. Which of the following is an effective variable name?

1 / 1 point

- ☐ `_able_int_variable`
- ☐ `able(variable)`
- ☒ `able_variable`
- ☐ `5_able_variables`

✓ Correct

7. In Python, which data type represents a sequence of characters and punctuation that contains textual information?

1 / 1 point

- ☒ String
- ☐ Boolean
- ☐ Integer
- ☐ Float

✓ Correct

8. A data professional converts floats into strings using a predefined function. This is an example of what type of conversion?

1 / 1 point

- ☐ Immutable conversion
- ☒ Explicit conversion
- ☐ Implicit conversion
- ☐ Mutable conversion

✓ Correct

9. You're working in a Jupyter notebook. Consider the following execution of cells:

1 / 1 point

Cell 1:

```
x = 10
```

Cell 2:

```
x = 'Python'
```

Cell 3:

```
x = 5
```

```
print(x)
```

If you run each cell in order and only one time, what will the output of cell 3 be?

- ☐ 'Python'
- ☒ 5
- ☐ 10
- ☐ x

✓ Correct

10. Which of the following are valid statements (i.e., will NOT result in a code error) in Python? Select all that apply.

1 / 1 point

☒ `6 > 3`

✓ Correct

☐ `15.56 \ x`

☒ `x == 2`

✓ Correct

☒ `14 % 2`

✓ Correct